DATA 442/642 · Advanced Machine Learning · Week 1

1.3 Derivatives for Optimization

Gradients point uphill. We walk the other way.

1. A derivative is a slope

Take a smooth map $J:\mathbb{R}\to\mathbb{R}$. A small step ε_x in the input makes a small step ε_y in the output. Near a point p, if J has no corners,

$$\varepsilon_y \approx a\,\varepsilon_x.$$

The number $a = J'(p)$ is the **derivative** at p.

- $a<0$: a small increase in x **decreases** J.
- $a>0$: a small increase in x **increases** J.
- $|a|$ is how fast that happens.

To **minimize** J, walk against the sign of a. That one sentence is gradient descent in one dimension.

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2. Continuity is not enough

Continuity says a small input change cannot jump the output. Smoothness says the graph has no kinks, so a linear approximation exists. Optimization algorithms in this course assume at least that much. ReLU networks are allowed later; they are smooth almost everywhere. You still write a gradient.

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3. Higher dimensions: the gradient

A function of two scalars, $J(x,y)=z$, is a surface. The analogue of a is the **gradient** ∇J: a vector of partial derivatives. It points to the direction of **steepest increase**. Descent uses $-\nabla J$.

Week 1 note **1.4** writes the gradient and the Hessian in the $\boldsymbol{\theta}$ notation of the rest of the course. Week 2 uses both.